

## Exercise 7: Summary Comparison

1.

| Program                | Collectives used                | Array allocation             | Manual summation loop at root? | Final result                         |
|------------------------|---------------------------------|------------------------------|--------------------------------|--------------------------------------|
| 1. Bcast + Send/Recv   | MPI_Bcast, MPI_Send, MPI_Recv   | Full array on every process  | Yes                            | Root only                            |
| 2. Scatter + Send/Recv | MPI_Scatter, MPI_Send, MPI_Recv | Only a chunk on each process | Yes                            | Root only                            |
| 3. Scatter + Gather    | MPI_Scatter, MPI_Gather         | Only a chunk on each process | Yes                            | Root only                            |
| 4. Scatter + Reduce    | MPI_Scatter, MPI_Reduce         | Only a chunk on each process | No                             | Root only                            |
| 5. Scatter + Allreduce | MPI_Scatter, MPI_Allreduce      | Only a chunk on each process | No                             | All processes                        |
| 6. Scatter + Scan      | MPI_Scatter, MPI_Scan           | Only a chunk on each process | No                             | Different cumulative result per rank |

2.

| Program             | 2 processes | 4 processes | 8 processes |
|---------------------|-------------|-------------|-------------|
| Bcast + Send/Recv   | 0.0046 sec  | 0.0375 sec  | 0.0230 sec  |
| Scatter + Send/Recv | 0.0042 sec  | 0.0024 sec  | 0.0018 sec  |
| Scatter + Gather    | 0.0034 sec  | 0.0018 sec  | 0.0016 sec  |
| Scatter + Reduce    | 0.0032 sec  | 0.0019 sec  | 0.0017 sec  |
| Scatter + Allreduce | 0.0028 sec  | 0.0020 sec  | 0.0017 sec  |
| Scatter + Scan      | 0.0065 sec  | 0.0051 sec  | 0.0053 sec  |